OOP Lab – Week 6

Question 1: Operator Overloading of <= for my_String

Write a C++ program to overload the relational operator <= to compare two objects of my_String class using a character array.

```
#include<iostream>
#include<cstring>
using namespace std;

class my_String{
    char str[50];

public:
    void input(){
        cout<<"Enter string: ";
        cin>>str;
    }

    bool operator <=(my_String s){
        return strcmp(str,s.str)<=0;
    }
};

int main(){
    my_String s1,s2;

    s1.input();
    s2.input();

    if(s1<=s2)
        cout<<"String 1 is less than or equal to String 2";
    else
        cout<<"String 1 is greater than String 2";

    return 0;
}
```

Question 2: Complex Number Addition and Subtraction

Write a C++ program to perform addition and subtraction on two complex numbers using operator overloading with member functions.

```
#include<iostream>
using namespace std;

class Complex{
    float real,imag;

public:
    Complex(float r=0,float i=0){
        real=r;
        imag=i;
    }

    void input(){
        cin>>real>>imag;
    }

    Complex operator +(Complex c){
        return Complex(real+c.real,imag+c.imag);
    }

    Complex operator -(Complex c){
        return Complex(real-c.real,imag-c.imag);
    }

    void display(){
        if(imag>=0)
            cout<<real<<"+"<<imag<<"i"<<endl;
        else
            cout<<real<<imag<<"i"<<endl;
    }
};

int main(){
    Complex c1,c2,sum,diff;

    cout<<"Enter real and imaginary parts of first complex number: ";
    c1.input();

    cout<<"Enter real and imaginary parts of second complex number: ";
    c2.input();

    sum=c1+c2;
    diff=c1-c2;

    cout<<"Addition: ";
    sum.display();

    cout<<"Subtraction: ";
    diff.display();

    return 0;
}
```